Splunk Detection Engineering and Threat Hunting Lab

Author: Eric Johnson

Date: June 2026

GitHub: https://github.com/ericjohnson4?tab=repositories

Executive Summary

This project demonstrates the deployment, configuration, and use of Splunk Enterprise as a Security Information and Event Management (SIEM) platform for detection engineering and threat hunting activities. A Windows 11 host was configured to forward Security, System, Application, and PowerShell Operational logs into Splunk for analysis.

Custom detections were developed to identify failed logon attempts (Event ID 4625), successful logons (Event ID 4624), new user account creation events (Event ID 4720), and PowerShell script execution activity (Event ID 4104). Splunk dashboards were created to visualize authentication activity, account management events, and PowerShell telemetry.

The lab demonstrates practical SOC analyst skills including log ingestion, data analysis, detection engineering, dashboard development, and threat hunting using real Windows event data.

Skills Demonstrated

- SIEM Administration
- Splunk Enterprise
- Detection Engineering
- Threat Hunting
- Windows Event Analysis
- PowerShell Logging
- Security Monitoring
- SPL Query Development
- Dashboard Development
- Security Operations (SOC)

Table of Contents

1. Lab Objectives

The objectives of this lab were to:

- Deploy and configure Splunk Enterprise.
- Create a dedicated index for security event collection.
- Ingest Windows Security, System, and Application event logs.
- Enable and collect PowerShell Script Block Logging events.
- Develop Splunk searches to identify security-relevant activity.
- Create dashboards for monitoring authentication, account management, and PowerShell execution activity.
- Demonstrate threat hunting and detection engineering capabilities using real-world Windows telemetry.

2. Lab Environment

Component	Details
SIEM Platform	Splunk Enterprise 10.4
Host Operating System	Windows 11 Pro
Index	security_lab
Data Sources	Security, System, Application Logs
Additional Logs	Microsoft-Windows-PowerShell/Operational
Detection Language	SPL (Search Processing Language)
Logging Features	PowerShell Script Block Logging

3. Splunk Deployment and Configuration

Figure 1

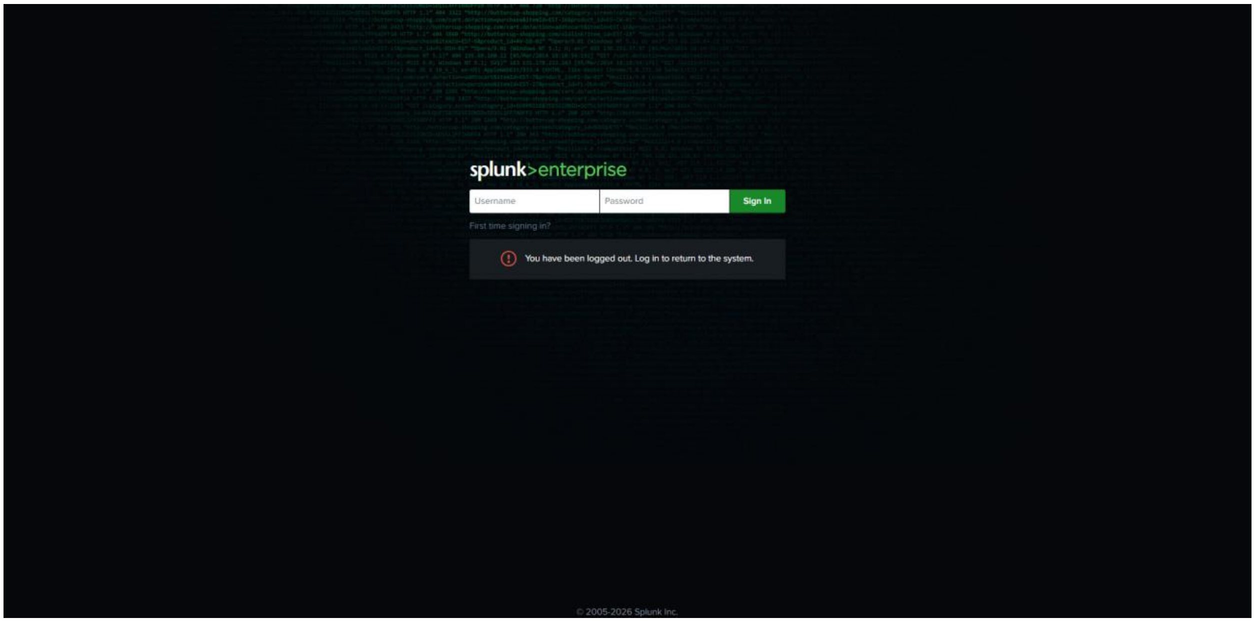


Figure 1 – Splunk Enterprise login portal

Figure 2

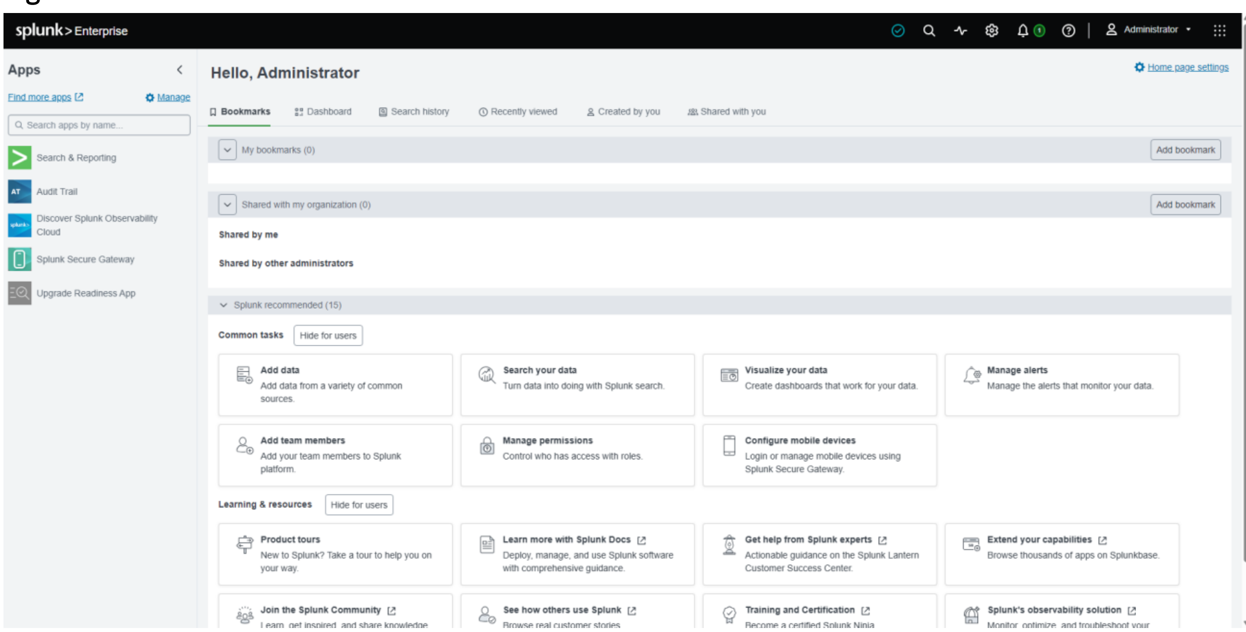


Figure 2 – Splunk Enterprise home dashboard after login

Figure 3

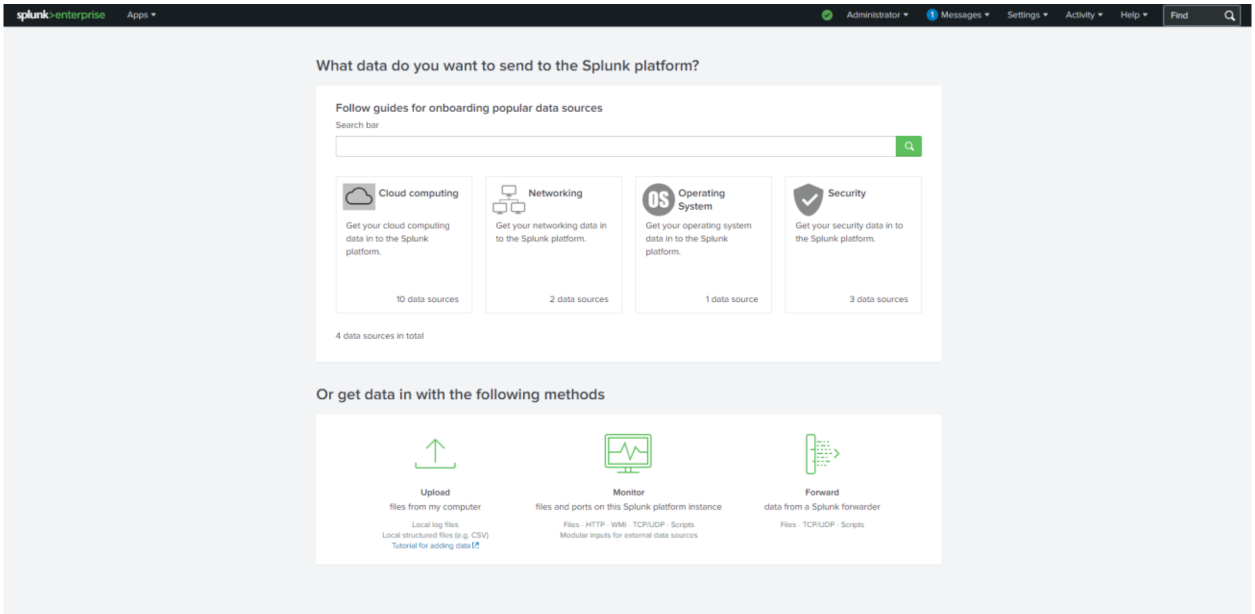


Figure 3 – Splunk settings menu used to access configuration options

Figure 4

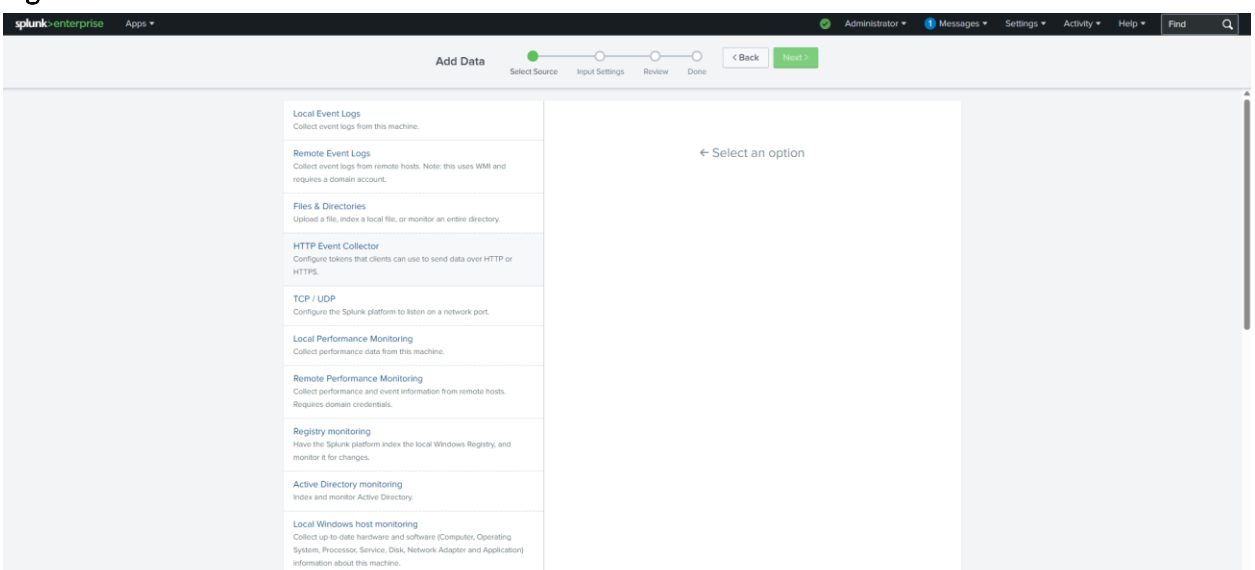


Figure 4 – Splunk Add Data workflow for configuring monitored inputs

Splunk Enterprise was installed and configured on a Windows 11 workstation. A dedicated index named security_lab was created to isolate security-related telemetry from default Splunk data sources.

Data inputs were configured to collect Security, System, and Application event logs. PowerShell Operational logs were later added through the inputs.conf configuration file to support script execution monitoring and threat hunting activities.

4. Windows Event Log Ingestion

Figure 5

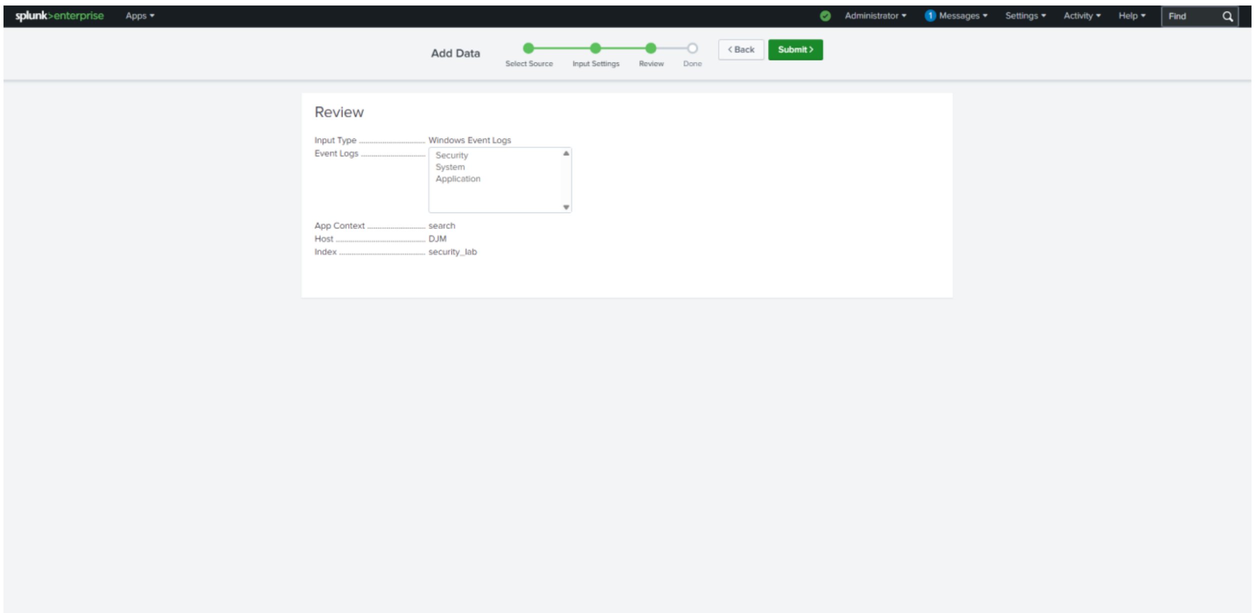


Figure 5 – Windows Event Log input configuration

Figure 6

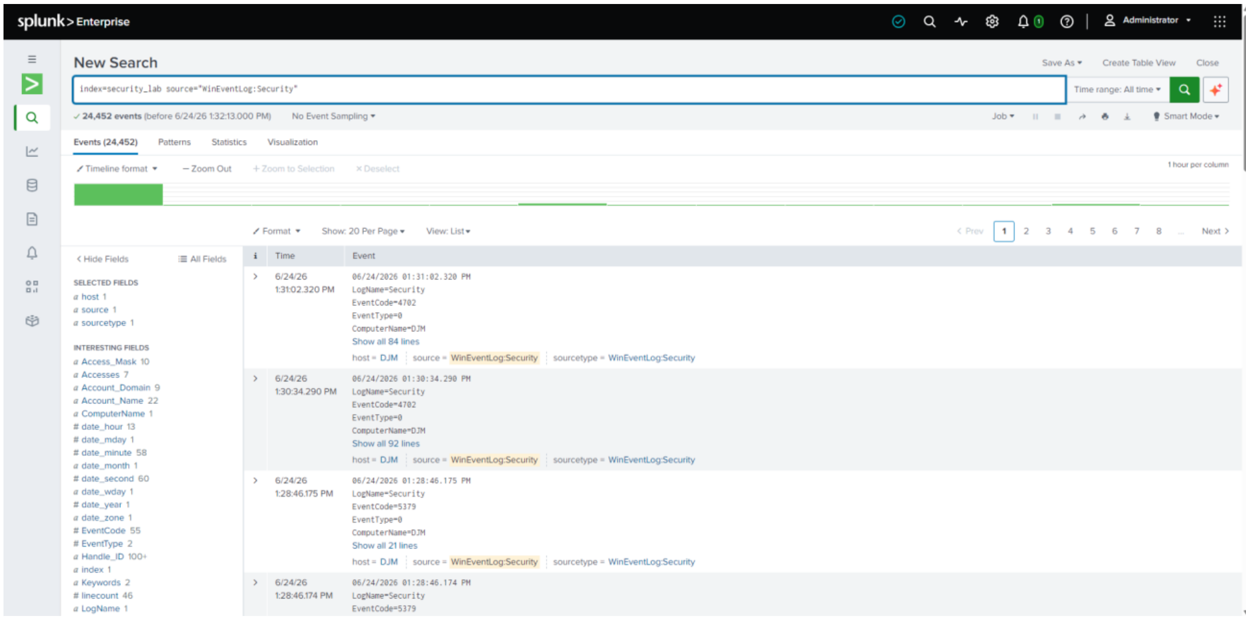


Figure 6 – Initial Windows Event Log ingestion verified in Splunk

Figure 7

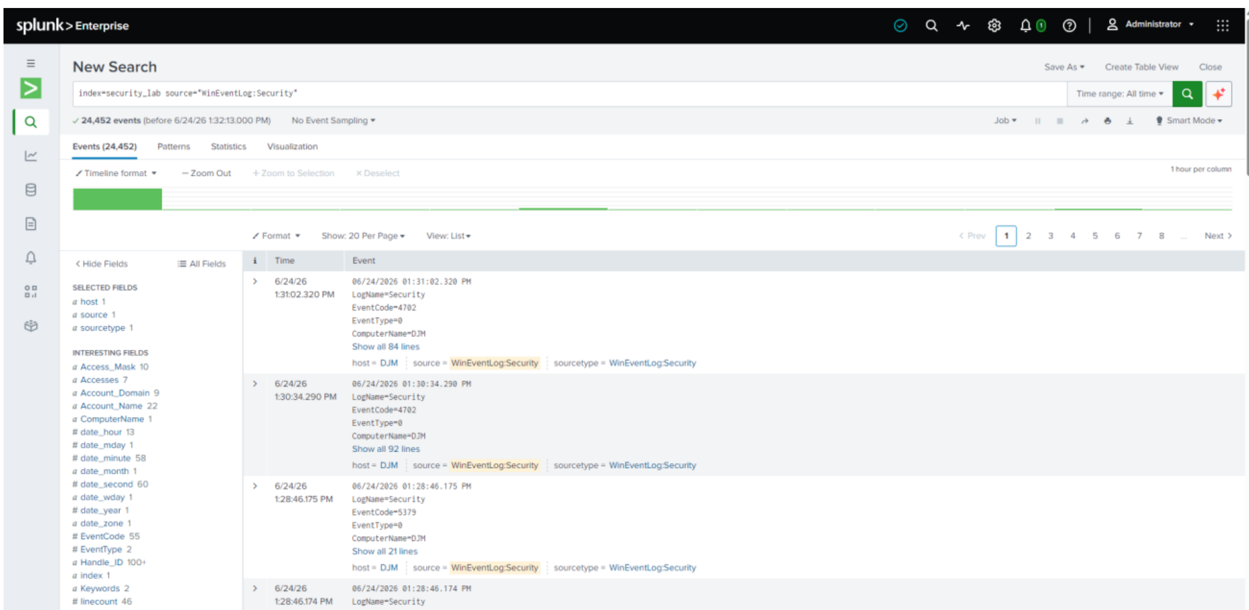


Figure 7 – Windows event fields and parsed log details in Splunk

5. Detection Engineering

Figure 8

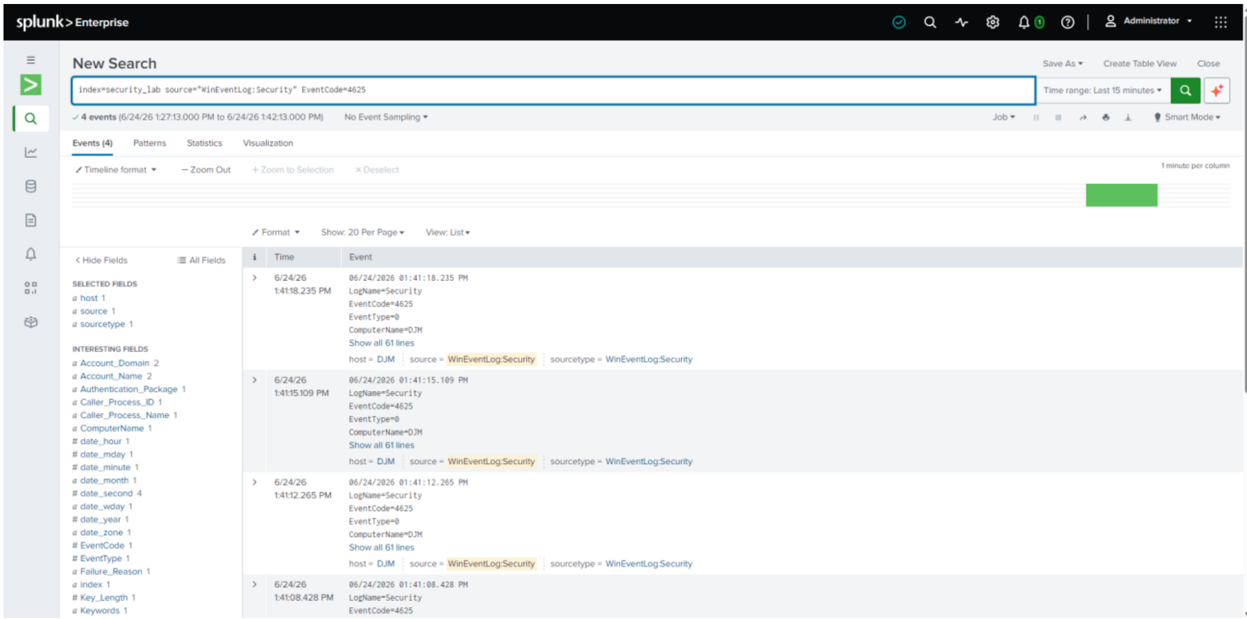


Figure 8 – Failed Logon Detection (Event ID 4625)

Figure 9

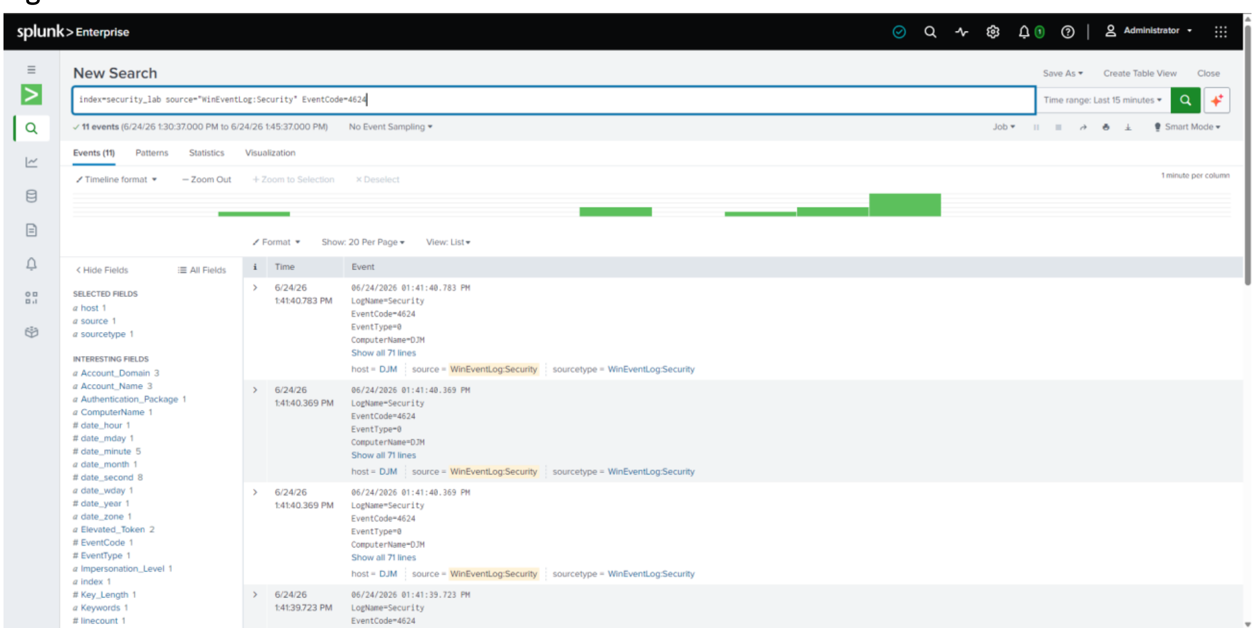


Figure 9 – Successful Logon Detection (Event ID 4624)

Figure 10

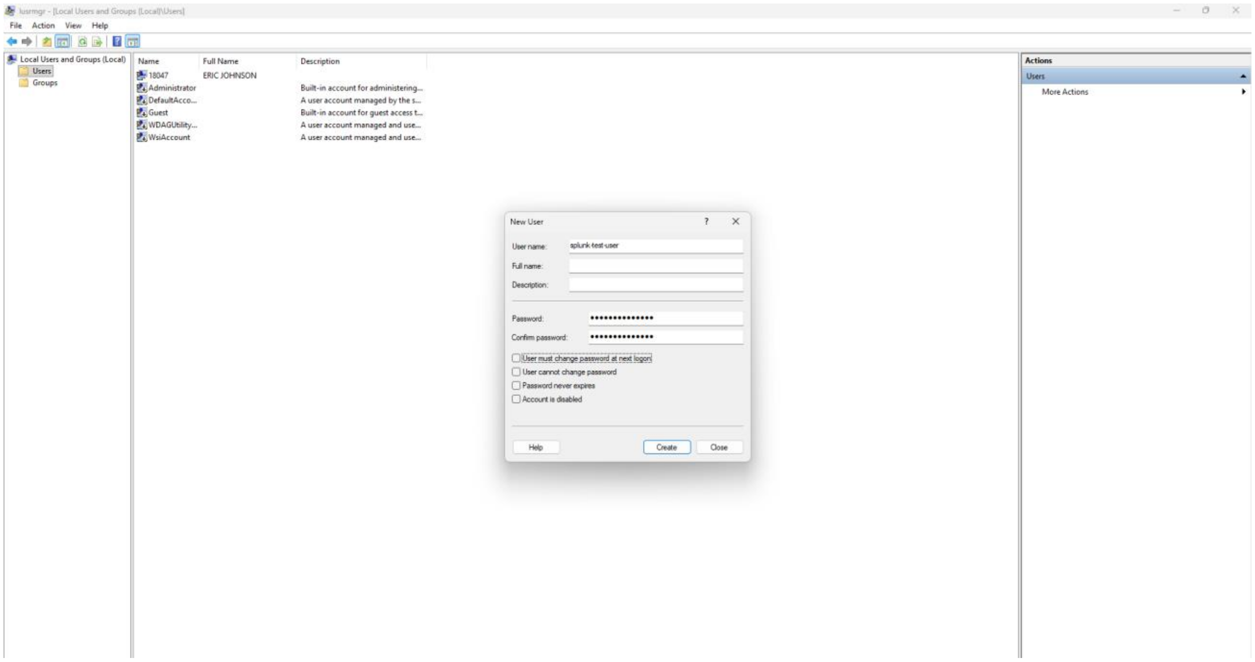


Figure 10 – Test User Creation Activity

Figure 11

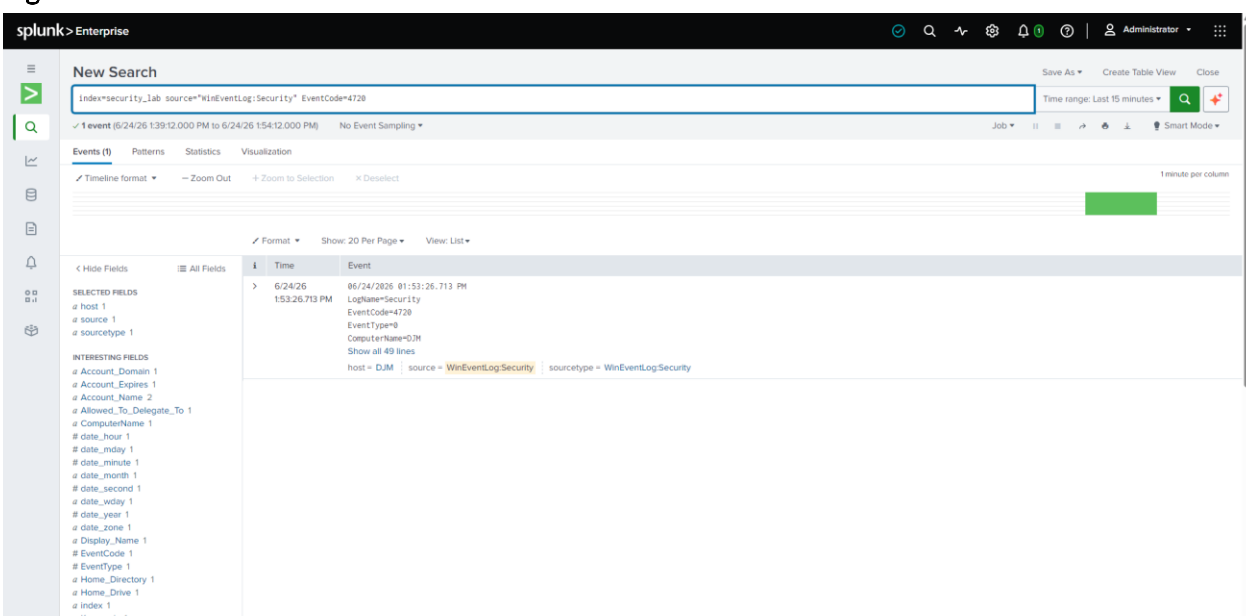


Figure 11 – New User Creation Detection (Event ID 4720)

The detection queries developed in this section demonstrate monitoring for authentication failures, successful logons, account creation activity, and administrative actions. These event types represent common security monitoring use cases within enterprise SOC environments.

6. PowerShell Logging Configuration and Threat Hunting

Figure 12

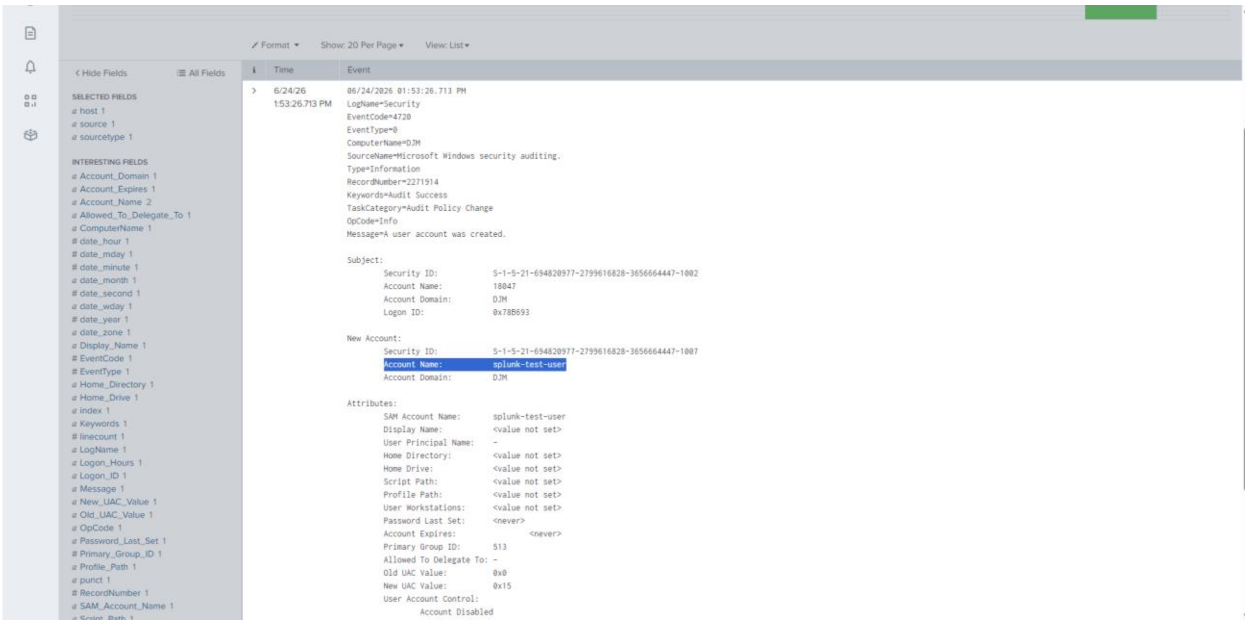
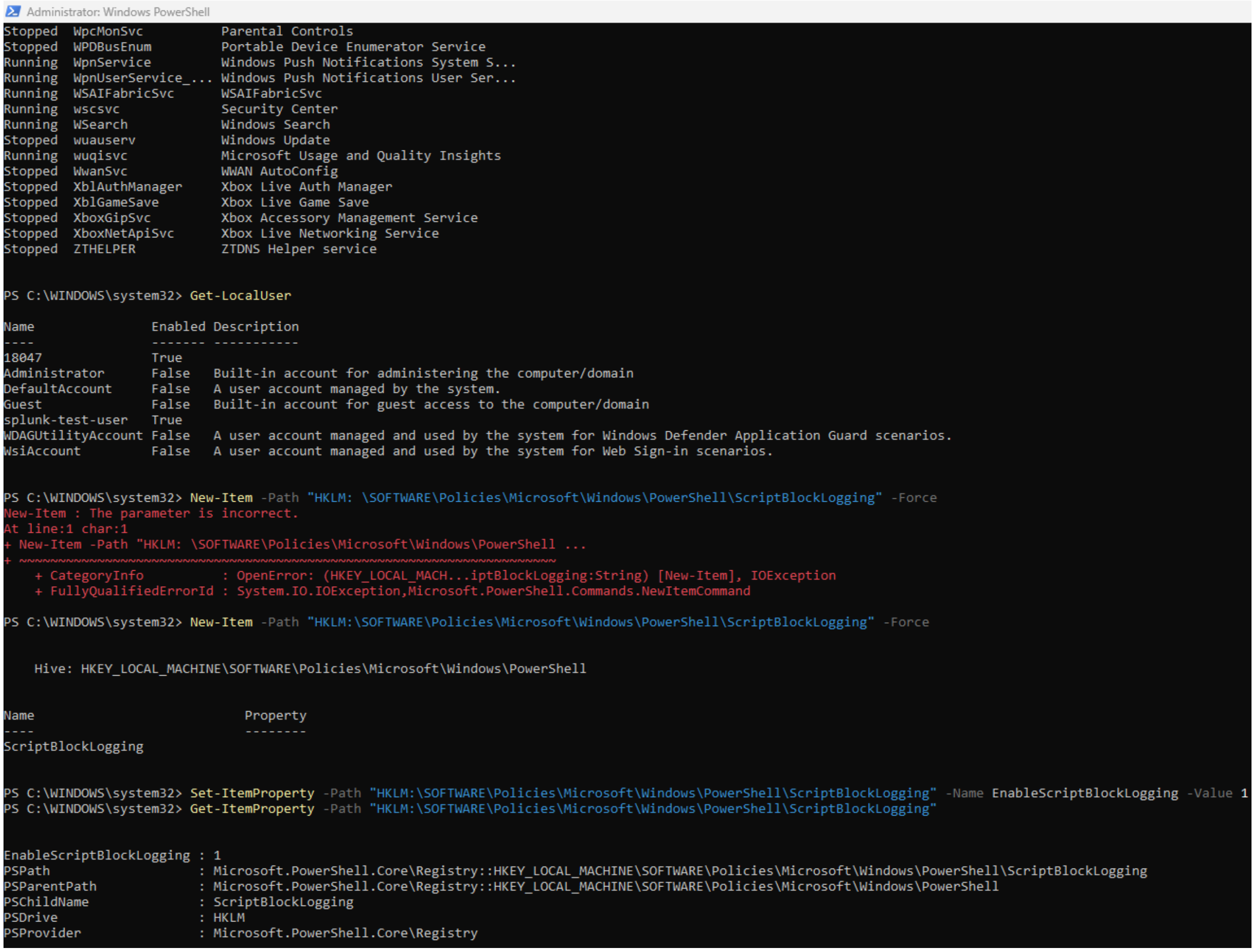


Figure 12 – Event details associated with Windows Event ID 4720

Figure 13



```
Administrator: Windows PowerShell
Stopped  WpcMonSvc          Parental Controls
Stopped  WPDBusEnum         Portable Device Enumerator Service
Running  WpnService         Windows Push Notifications System S...
Running  WpnUserService_... Windows Push Notifications User Ser...
Running  WSAIFabricSvc      WSAIFabricSvc
Running  wscsvc             Security Center
Running  WSearch            Windows Search
Stopped  wuauserv           Windows Update
Running  wuqisvc            Microsoft Usage and Quality Insights
Stopped  WwanSvc            WWAN AutoConfig
Stopped  XblAuthManager     Xbox Live Auth Manager
Stopped  XblGameSave        Xbox Live Game Save
Stopped  XboxGipSvc         Xbox Accessory Management Service
Stopped  XboxNetApiSvc      Xbox Live Networking Service
Stopped  ZTHELPER           ZTDNS Helper service


PS C:\WINDOWS\system32> Get-LocalUser

Name               Enabled Description
----               ------- -----------
18047              True
Administrator      False   Built-in account for administering the computer/domain
DefaultAccount     False   A user account managed by the system.
Guest              False   Built-in account for guest access to the computer/domain
splunk-test-user   True
WDAGUtilityAccount False   A user account managed and used by the system for Windows Defender Application Guard scenarios.
WsiAccount         False   A user account managed and used by the system for Web Sign-in scenarios.


PS C:\WINDOWS\system32> New-Item -Path "HKLM: \SOFTWARE\Policies\Microsoft\Windows\PowerShell\ScriptBlockLogging" -Force
New-Item : The parameter is incorrect.
At line:1 char:1
+ New-Item -Path "HKLM: \SOFTWARE\Policies\Microsoft\Windows\PowerShell ...
+ ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : OpenError: (HKEY_LOCAL_MACH...iptBlockLogging:String) [New-Item], IOException
    + FullyQualifiedErrorId : System.IO.IOException,Microsoft.PowerShell.Commands.NewItemCommand

PS C:\WINDOWS\system32> New-Item -Path "HKLM:\SOFTWARE\Policies\Microsoft\Windows\PowerShell\ScriptBlockLogging" -Force


    Hive: HKEY_LOCAL_MACHINE\SOFTWARE\Policies\Microsoft\Windows\PowerShell


Name                           Property
----                           --------
ScriptBlockLogging


PS C:\WINDOWS\system32> Set-ItemProperty -Path "HKLM:\SOFTWARE\Policies\Microsoft\Windows\PowerShell\ScriptBlockLogging" -Name EnableScriptBlockLogging -Value 1
PS C:\WINDOWS\system32> Get-ItemProperty -Path "HKLM:\SOFTWARE\Policies\Microsoft\Windows\PowerShell\ScriptBlockLogging"


EnableScriptBlockLogging : 1
PSPath                   : Microsoft.PowerShell.Core\Registry::HKEY_LOCAL_MACHINE\SOFTWARE\Policies\Microsoft\Windows\PowerShell\ScriptBlockLogging
PSParentPath             : Microsoft.PowerShell.Core\Registry::HKEY_LOCAL_MACHINE\SOFTWARE\Policies\Microsoft\Windows\PowerShell
PSChildName              : ScriptBlockLogging
PSDrive                  : HKLM
PSProvider               : Microsoft.PowerShell.Core\Registry
```

Figure 13 – PowerShell Script Block Logging enabled through Group Policy

Figure 14

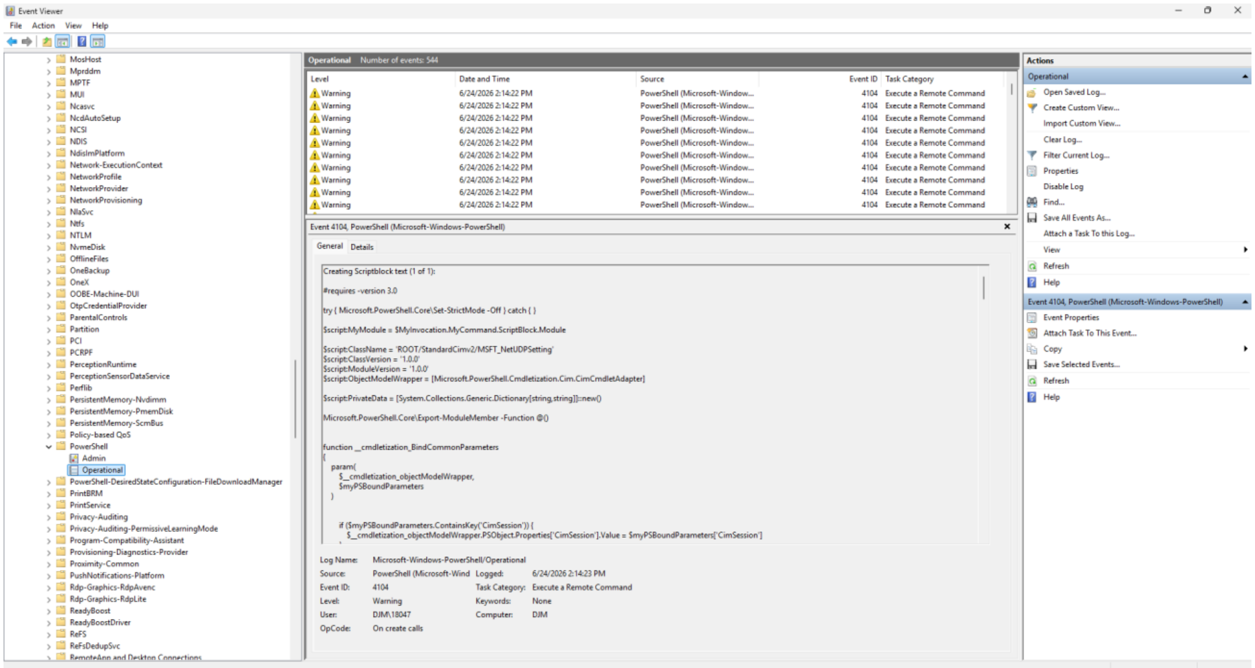


Figure 14 – Verification of Event ID 4104 in Windows Event Viewer

Figure 15

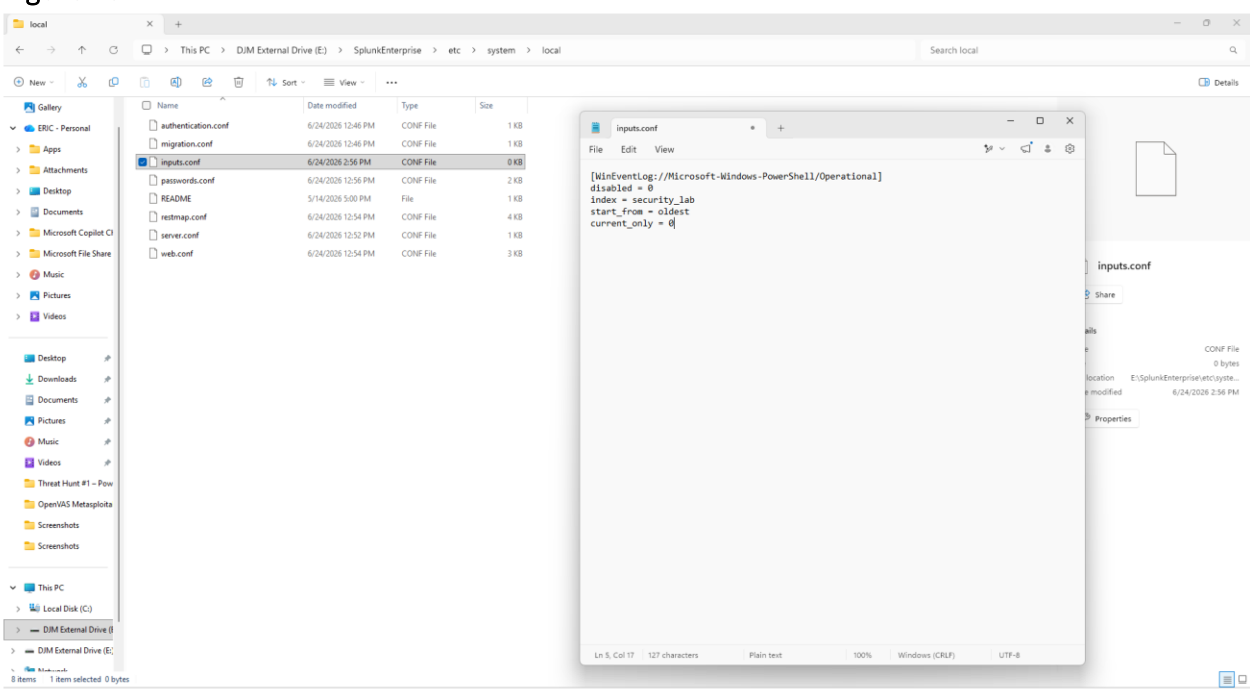


Figure 15 – Configuration of PowerShell Operational log collection in Splunk

Figure 16

inputs.conf	6/24/2026 3:02 PM	CONF File	1 KB

Figure 16 – PowerShell input configuration saved and validated

Figure 17

```
PS C:\WINDOWS\system32> cd E:\SplunkEnterprise\bin
PS E:\SplunkEnterprise\bin> .\splunk.exe restart
Splunkd: Stopped

Splunk> Map. Reduce. Recycle.

Checking prerequisites...
        Checking http port [8000]: open
        Checking mgmt port [8089]: open
        Checking appserver port [127.0.0.1:8065]: open
        Checking kvstore port [8191]: open
        Checking configuration... Done.
New certs have been generated in 'E:\SplunkEnterprise\etc\auth'.
        Checking critical directories...        Done
        Checking indexes...
                (skipping validation of index paths because not running as NT SERVICE\Splunkd)
                Validated: _audit _configtracker _dm_summary _dsappevent _dsclient _dsphonehome _internal _introspection _metrics _metrics_rollup _telemetry _thefishbucket history main security_lab summary
        Done
        Checking filesystem compatibility...  Done
        Checking conf files for problems...
        Done
        Checking default conf files for edits...
        Validating installed files against hashes from 'E:\SplunkEnterprise\splunk-10.4.0-f798d4d49089-windows-x64-manifest'
        All installed files intact.
        Done
All preliminary checks passed.

Starting splunk server daemon (splunkd)...

Splunkd: Starting (pid 3424)
Done


Waiting for web server at http://127.0.0.1:8000 to be available... Done


If you get stuck, we're here to help.
Look for answers here: http://docs.splunk.com

The Splunk web interface is at http://DJM:8000

PS E:\SplunkEnterprise\bin> _
```

Figure 17 – Splunk service restarted after PowerShell log configuration changes

Figure 18

```
PS E:\SplunkEnterprise\bin> E:\SplunkEnterprise\bin\splunk.exe btool inputs list "WinEventLog://Microsoft-Windows-PowerShell/Operational" --debug
E:\SplunkEnterprise\etc\system\local\inputs.conf   [WinEventLog://Microsoft-Windows-PowerShell/Operational]
E:\SplunkEnterprise\etc\system\local\inputs.conf   current_only = 0
E:\SplunkEnterprise\etc\system\local\inputs.conf   disabled = 0
E:\SplunkEnterprise\etc\system\default\inputs.conf evt_dc_name =
E:\SplunkEnterprise\etc\system\default\inputs.conf evt_dns_name =
E:\SplunkEnterprise\etc\system\default\inputs.conf evt_resolve_ad_obj = 0
host = $decideOnStartup
E:\SplunkEnterprise\etc\system\local\inputs.conf   index = security_lab
E:\SplunkEnterprise\etc\system\default\inputs.conf interval = 60
E:\SplunkEnterprise\etc\system\local\inputs.conf   start_from = oldest
```

Figure 18 – Verification of active PowerShell log input using Splunk btool

Figure 19

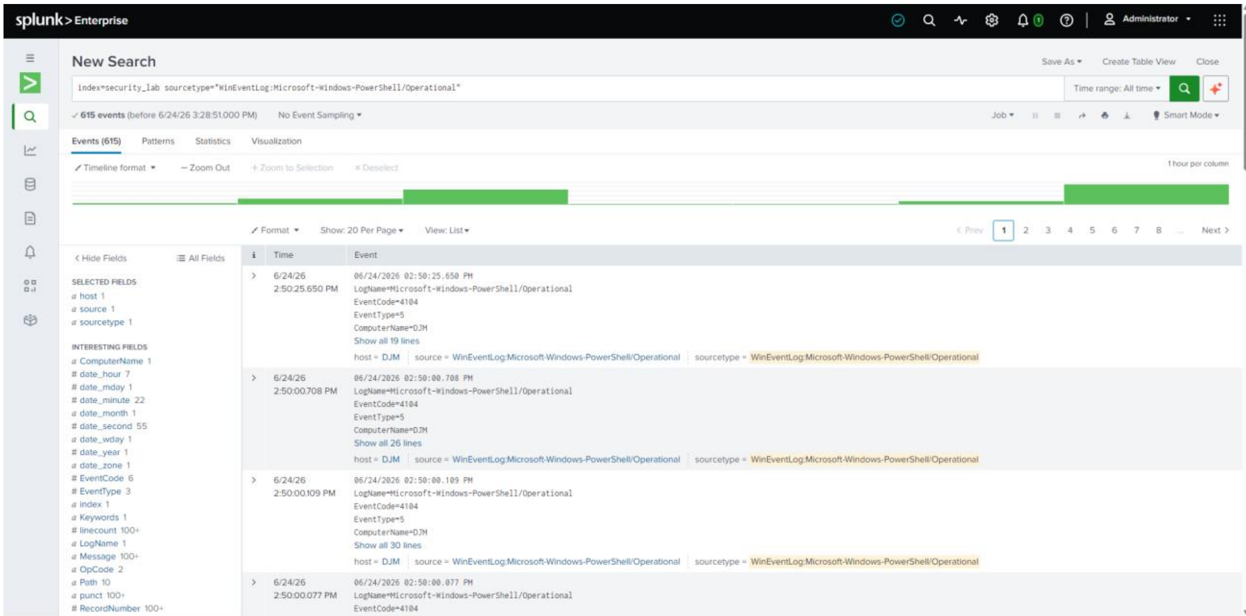


Figure 19 – PowerShell Script Block Logging events detected in Splunk

Figure 20

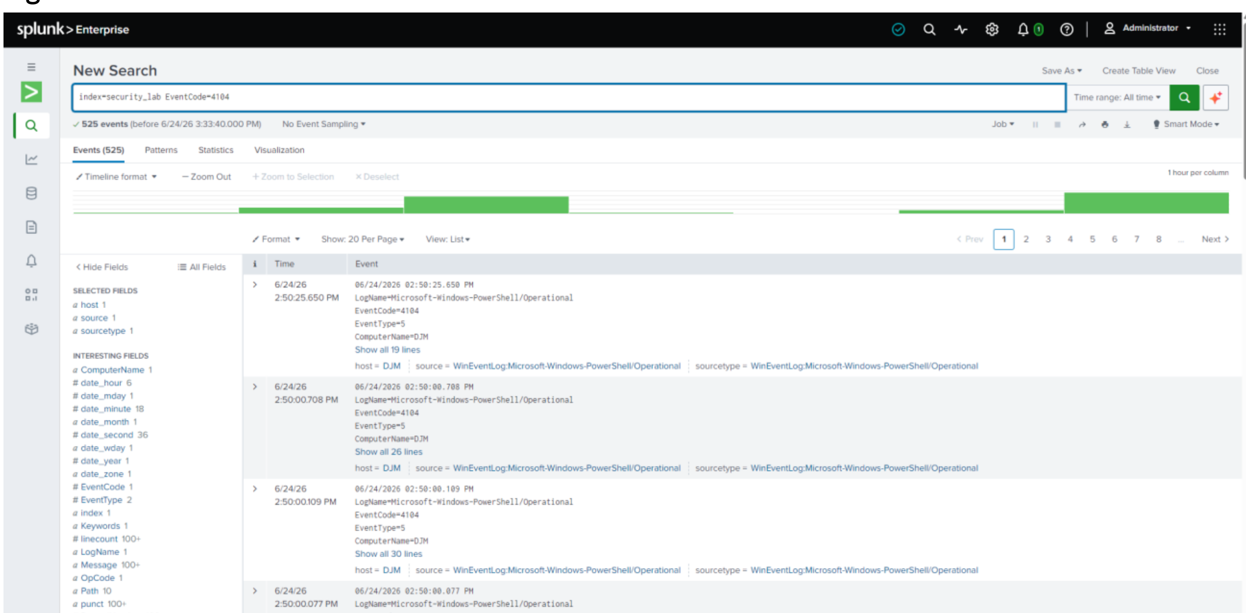


Figure 20 – SPL query used to identify PowerShell Event ID 4104 activity

PowerShell Script Block Logging was enabled to provide detailed visibility into PowerShell execution activity. Event ID 4104 records the contents of PowerShell script blocks, making it valuable for threat hunting and detection engineering.

The Microsoft-Windows-PowerShell/Operational log was added to Splunk through inputs.conf, validated using Splunk btool, and activated through a service restart. Verification searches confirmed successful ingestion of PowerShell telemetry into the security_lab index.

Custom SPL searches were then developed to identify and investigate PowerShell execution activity within Splunk.

7. Dashboard Development

Figure 21

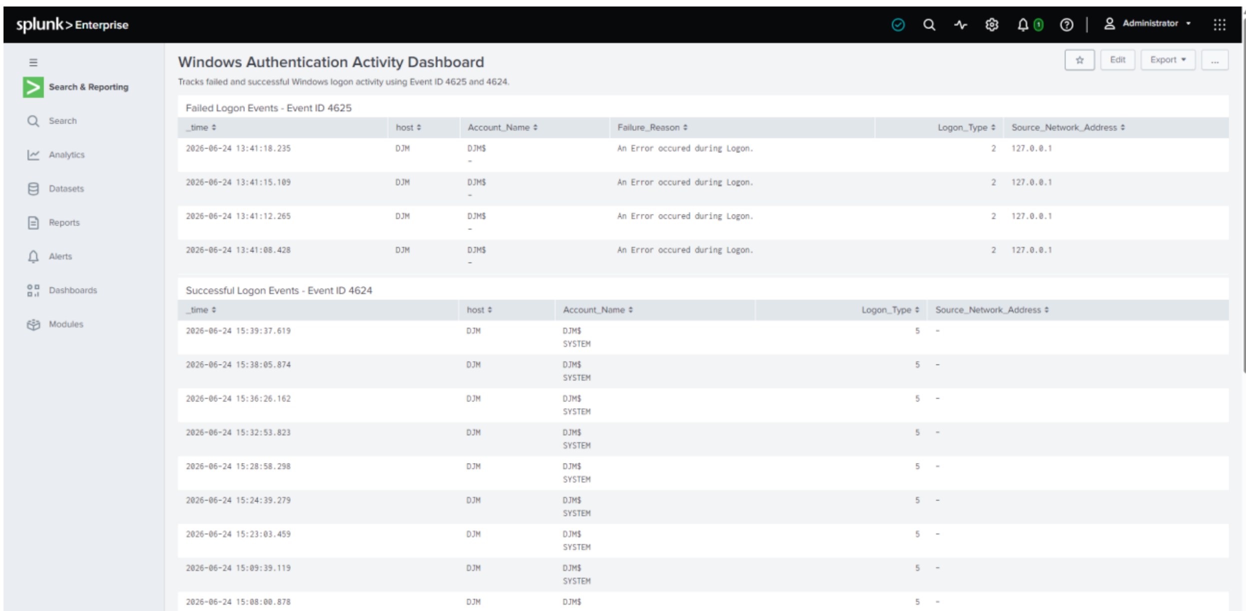


Figure 21 – Dashboard displaying failed and successful Windows authentication activity

Figure 22

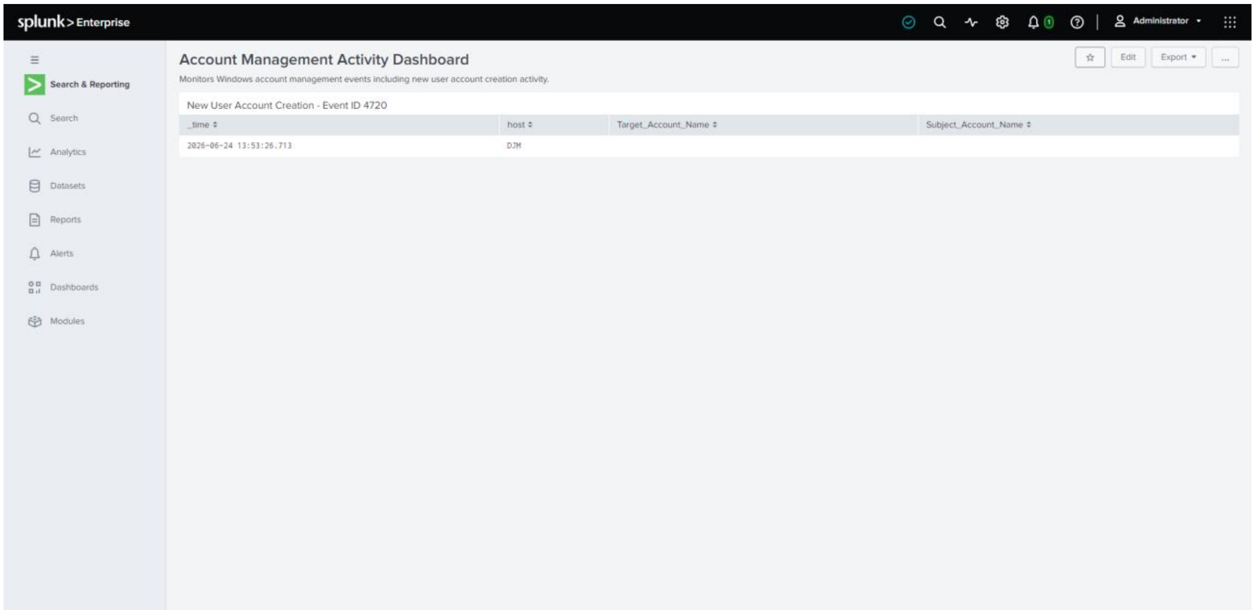


Figure 22 – Dashboard monitoring Windows user account creation activity

Figure 23

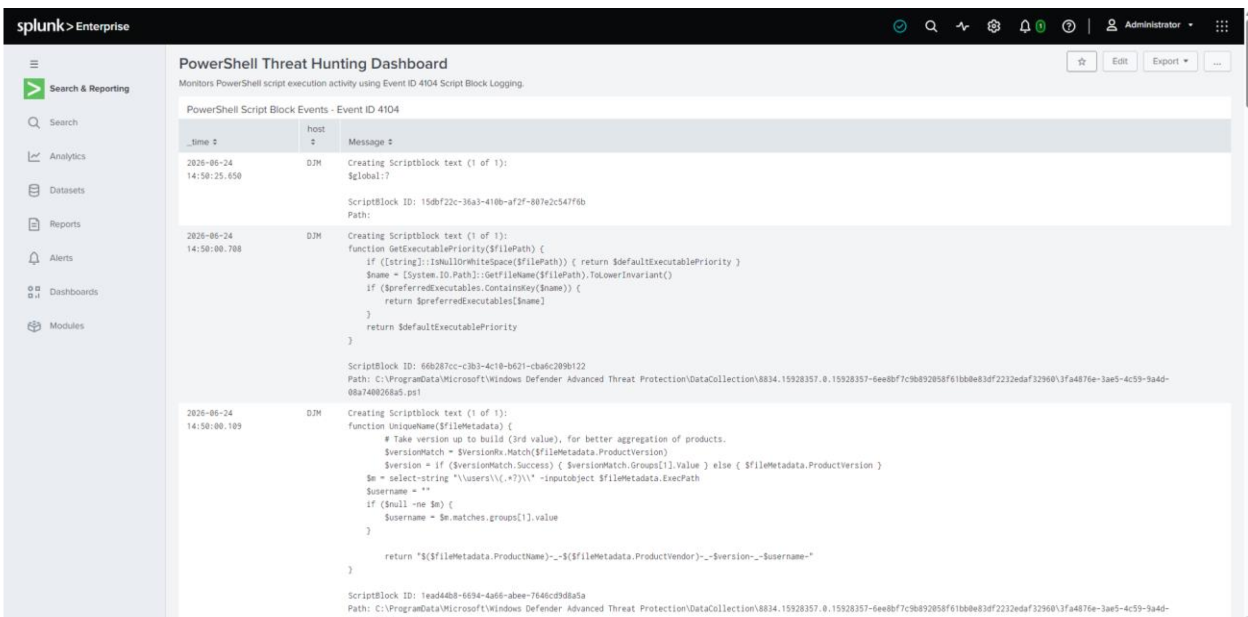


Figure 23 – Dashboard displaying PowerShell Script Block Logging events for threat hunting

Custom dashboards were developed to transform raw Windows event data into actionable security monitoring views. Authentication dashboards provided visibility into successful and failed logon activity, while account management dashboards monitored user creation events. A dedicated PowerShell dashboard was created to support threat hunting and analysis of script execution activity.

These dashboards demonstrate how Splunk can be used to centralize, visualize, and investigate security-relevant events within a SIEM platform. Dashboard development improves analyst efficiency by providing rapid access to key security indicators and detection results.

8. Findings and Analysis

The lab successfully demonstrated end-to-end security monitoring using Splunk Enterprise. Windows Security, System, Application, and PowerShell Operational logs were ingested into a dedicated security_lab index and analyzed using SPL queries.

Detection engineering activities successfully identified failed logon attempts (Event ID 4625), successful logon activity (Event ID 4624), new user account creation events (Event ID 4720), and PowerShell script execution activity (Event ID 4104). PowerShell Script Block Logging provided detailed visibility into script execution, enhancing threat hunting capabilities.

The creation of dashboards improved visibility into authentication activity, account management events, and PowerShell telemetry, demonstrating how SIEM technologies can support security operations and incident investigation workflows.

9. Lessons Learned

Key lessons learned from this project include:

- Successful SIEM deployments require proper log source configuration and validation.
- Windows Event IDs provide valuable telemetry for authentication monitoring, account management detection, and threat hunting activities.
- PowerShell Script Block Logging significantly improves visibility into script execution behavior.
- SPL enables rapid development of custom detections and investigative searches.
- Dashboards improve analyst efficiency by transforming large volumes of log data into actionable visualizations.
- Detection engineering and threat hunting depend on understanding both event data and log collection architecture.

10. Conclusion

This project successfully demonstrated the deployment, configuration, and operational use of Splunk Enterprise for detection engineering and threat hunting. Multiple Windows log sources were collected and analyzed, custom detections were developed, and dashboards were created to support security monitoring activities.

Through this lab, practical experience was gained in SIEM administration, Windows event analysis, SPL query development, dashboard creation, and threat hunting workflows. These activities reflect many of the responsibilities performed by SOC analysts and cybersecurity professionals in modern security operations environments.

References

Microsoft. (2025). Windows Security Auditing Event Reference.

https://learn.microsoft.com/en-us/windows/security/threat-protection/auditing/

Microsoft. (2025). PowerShell Logging and Script Block Logging.

https://learn.microsoft.com/en-us/powershell/

Splunk. (2025). Splunk Enterprise Documentation.

https://docs.splunk.com

Splunk. (2025). Search Processing Language (SPL) Reference.

https://docs.splunk.com/Documentation/Splunk/latest/SearchReference